Cyrus Guided Notes: Stanford CS231n Deep Learning for Computer Vision

Stanford course pack

2026-05-15

Course source

Official source: `https://cs231n.stanford.edu/`

Why this course is in the system. CNNs, optimization, backpropagation, visual representation, and spatial-intelligence transfer.

Learning output. Already connected to guided PDF, notes index, and assignment links.

Prerequisite checkpoint

- Linear algebra
- Calculus
- Python and NumPy

Formula checkpoint

Do not memorize these first. Read each formula as a sentence: what changes, what is observed, what is optimized, and what output it produces.

$$p_k = \frac{e^{s_k}}{\sum_j e^{s_j}}$$
$$L = -\log p_y$$
$$(I * K)_{i,j} = \sum_{m,n} I_{i+m,j+n} K_{m,n}$$
$$\frac{\partial L}{\partial W} = \frac{\partial L}{\partial s} \frac{\partial s}{\partial W}$$

Visual page

Draw pixels -> convolution -> activation -> class score -> loss.

GoodNotes pages

- Page ST-DL001: softmax and cross entropy
- Page ST-DL002: convolution
- Page ST-DL003: backprop

Web interaction

A small convolution kernel visual that updates feature responses.

Obsidian and Notion

Layer	Output
Obsidian	Stanford Spatial AI -> CS231n
Notion	Assignment and formula-progress status.

First study move

Open the official source, copy the prerequisite checkpoint into GoodNotes, then write one formula in your own words before watching or reading more.